# Ching-Hsiang (Adam) Chang

## Data Engineer

Taipei, Taiwan lavouver96@gmail.com (+886) 922 554 637 linkedin.com/in/ching-hsiang-chang-782281217
github.com/AdamXiang adam-website-tau.vercel.app

### Summary

Data engineer at Cathay Life Insurance with hands-on experience across insurance, consulting, market research and consumer hardware. I build automated pipelines, medallion data models and dashboards with Python, SQL, dbt, Databricks and Azure. Recent results: a DataStage-to-dbt migration that cut deployment and maintenance time by over 50%, a 40-minute ETL job brought under 1 minute per file, and 30–80× faster database ingestion.

### Experience

**Data Engineer,** Cathay Life Insurance — May 2026 – Present

- Designed scalable ETL pipelines with IBM DataStage on a medallion architecture, implementing change data capture and SCD Type 2; refactored legacy workflows for a 30% gain in overall run efficiency.
- Drove the migration of pipelines from DataStage to dbt, simplifying the system and adding version control, automated testing and data lineage, which cut deployment and maintenance time by more than 50%.

**Data Engineer,** DATARGET — Nov 2025 – Apr 2026

- Led the move from Pandas to parallel processing with ClickHouse Local, bypassing the Python GIL and cutting ETL time from 40 minutes to under 1 minute per file (99% faster).
- Replaced standard SQL inserts with MSSQL BCP for 30–80× faster ingestion; built a Python HandlerFactory (OOP, abstract base classes) for automated schema generation.
- Designed the migration strategy from Azure to Microsoft Fabric and introduced Airflow and dbt with shared code standards.
- Diagnosed geospatial precision errors (latitude/longitude rounding) and time-series logic conflicts, recovering 25× more valid data points for forecasting models.
- Presented a Superset vs. Metabase trade-off analysis to leadership and deployed the chosen tool in Docker for geospatial analytics.

**Go-to-Market Intern,** Logitech — Jul 2024 – Aug 2024

- Built an automated Tableau dashboard for real-time product metrics, and a smart Google Sheets form that removed manual entry from the design workflow.

**Data Analyst Intern,** uppeta — Apr 2024 – Jun 2024

- Consulted on 5+ client projects to improve user registration, and automated internal data processing with Google Apps Script.

**Data Engineer Intern,** CXG — Jul 2023 – Apr 2024

- Engineered a real-time Power BI and Azure Synapse reporting solution with operations managers, reducing reporting time by 75%.
- Architected and standardised a medallion architecture (bronze, silver, gold) as the single source of truth for analysts.
- Orchestrated ELT and HTTP API ingestion pipelines in Azure Data Factory, ensuring 100% on-time scheduled refreshes.
- Migrated legacy Excel workflows to Metabase with optimised SQL transformations on Azure Synapse.

**Data Analyst Intern,** Cathay Financial Holdings — Feb 2023 – Jul 2023

- Designed a FHIR-standard ER schema that reduced inter-hospital data exchange latency by 75%.
- Built an end-to-end medical LINE bot (Flask, MongoDB) with automated scraping pipelines; designed ChatGPT prompt workflows to automate XML-to-JSON conversion of legacy records.

### Selected projects github.com/AdamXiang

- **AdmGPT**: self-hosted ChatGPT-style assistant on Claude and LangGraph with streaming chat, RAG over uploaded documents, web search and long-term memory.
- **AI Agent Evaluation**: end-to-end evaluation of a tool-using refund agent with DeepEval across answer, tool-call and trace-level metrics.
- **Banking CDC pipeline**: PostgreSQL → Debezium/Kafka → MinIO → Snowflake with SCD Type 2 marts, orchestrated by Airflow and dbt.
- **Kitchen Clash** (kitchen-clash.onrender.com): browser pickleball game with online multiplayer and voice chat.

### Skills

| | |
|---|---|
| **Languages** | Python, SQL, PySpark |
| **Data engineering** | dbt, Apache Airflow, IBM DataStage, Azure Data Factory, Kafka, Spark, Flink, Delta Live Tables, medallion architecture, CDC |
| **Platforms** | Databricks, Azure Synapse, Microsoft Fabric, Snowflake, ClickHouse, SQL Server, MongoDB, Docker |
| **Analytics** | Power BI, Tableau, Metabase, Apache Superset, Google Sheets, Excel |
| **AI engineering** | LangGraph, LangChain, RAG, Model Context Protocol, DeepEval |

### Education

**M.S., Brain Science** — 2021 – 2024
National Yang Ming Chiao Tung University

**B.S., Animal Science** — 2017 – 2021
National Taiwan University

### Certification and service

**Databricks Certified Data Engineer Associate**

**Diplomatic Alternative Service**, Taiwan Technical Mission in Eswatini, Nov 2024 – Jul 2025. Supported a national pig industry improvement project.